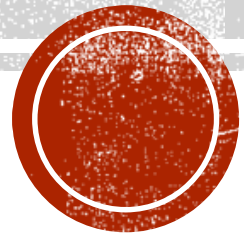
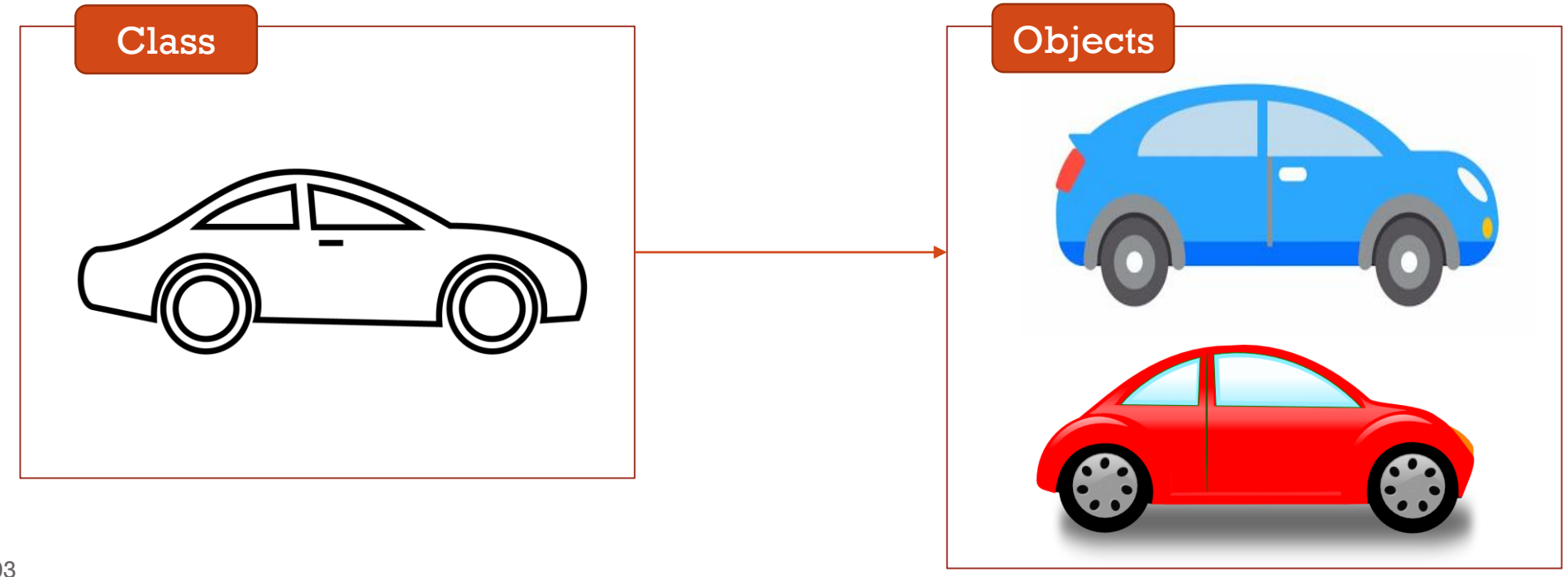
OBJECT ORIENTED PROGRAMMING

Working with Classes …

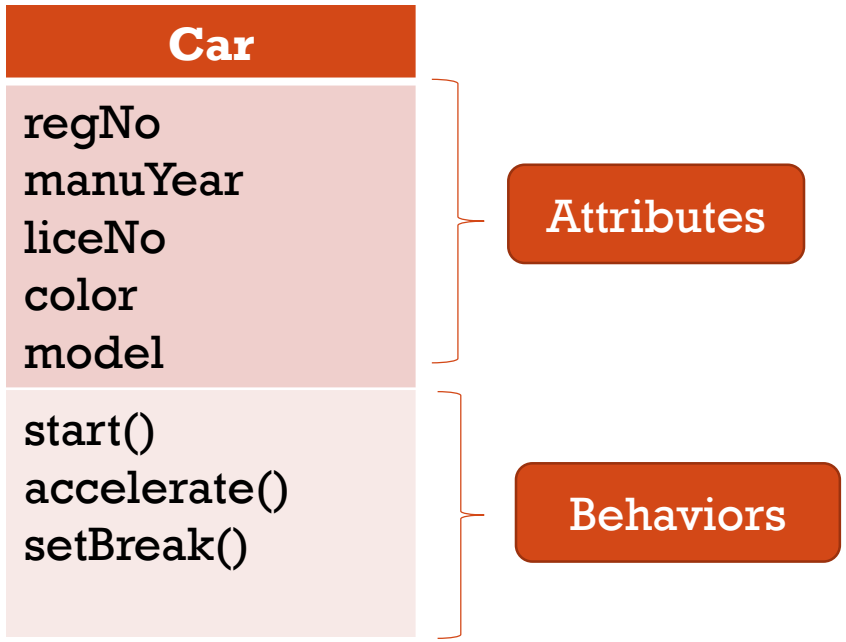


WORKING WITH CLASSES

- Two main building blocks of Object Orientation.
- A class is a **template** for objects, and an object is an **instance** of a class.



CLASSES IN C++




```
class car
{
public:
        int regNo;
        int manuYear;
        string liceNo;
        string color;
        string model;

        void start();
        int accelerate();
        void setBreak();
};
```

Class Body

Access specifier

Data Members

Member Functions

End of the class Definition

CLASS MEMBERS

Functions that are declared within a class are called **member functions**

Variables that are declared within a class are called **data members**

DECLARE OBJECTS

Syntax:

```
ClassName ObjectName;
```

DOT OPERATOR

- To refer to a member of a class outside the class definition: Prefix the member's name with the dot operator and the name of the object it belongs to.
- The member may be either data or a function.

```
objectName.memberName
```

CREATING OBJECTS

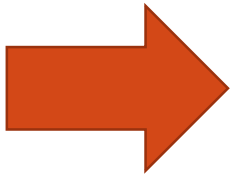
- An object is created from a class.
- To create an object, specify the class name, followed by the object name.
  - Example: car myRedCar;
- To access the class attributes/member functions, use the dot syntax (.) on the object.
  - Example:

    myRedCar.regNo;

```
Int main()
{
car myRedCar;
car myBlueCar={456,2018,"CP345","Blue","B"};
}
```

ACTIVITY

```
1   #include<iostream>
2   using namespace std;
3
4   class car
5   {
6       int regNo;
7       int manuYear;
8       string liceNo;
9       string color;
10      string model;
11
12      void start();
13      int accelerate();
14      void setbreak();
15  };
16
17  int main()
18  {
19      car myRedCar ={234,2017,"Wp4567","Red","C"};
20          car myBlueCar={456,2018,"CP345","Blue","B"};
21
22      cout<<"My Red Car: "<<myRedCar.liceNo<<" "<<endl;
23      cout<<"My Blue Car: " <<myBlueCar.liceNo<<" "<<endl;
24  }
```

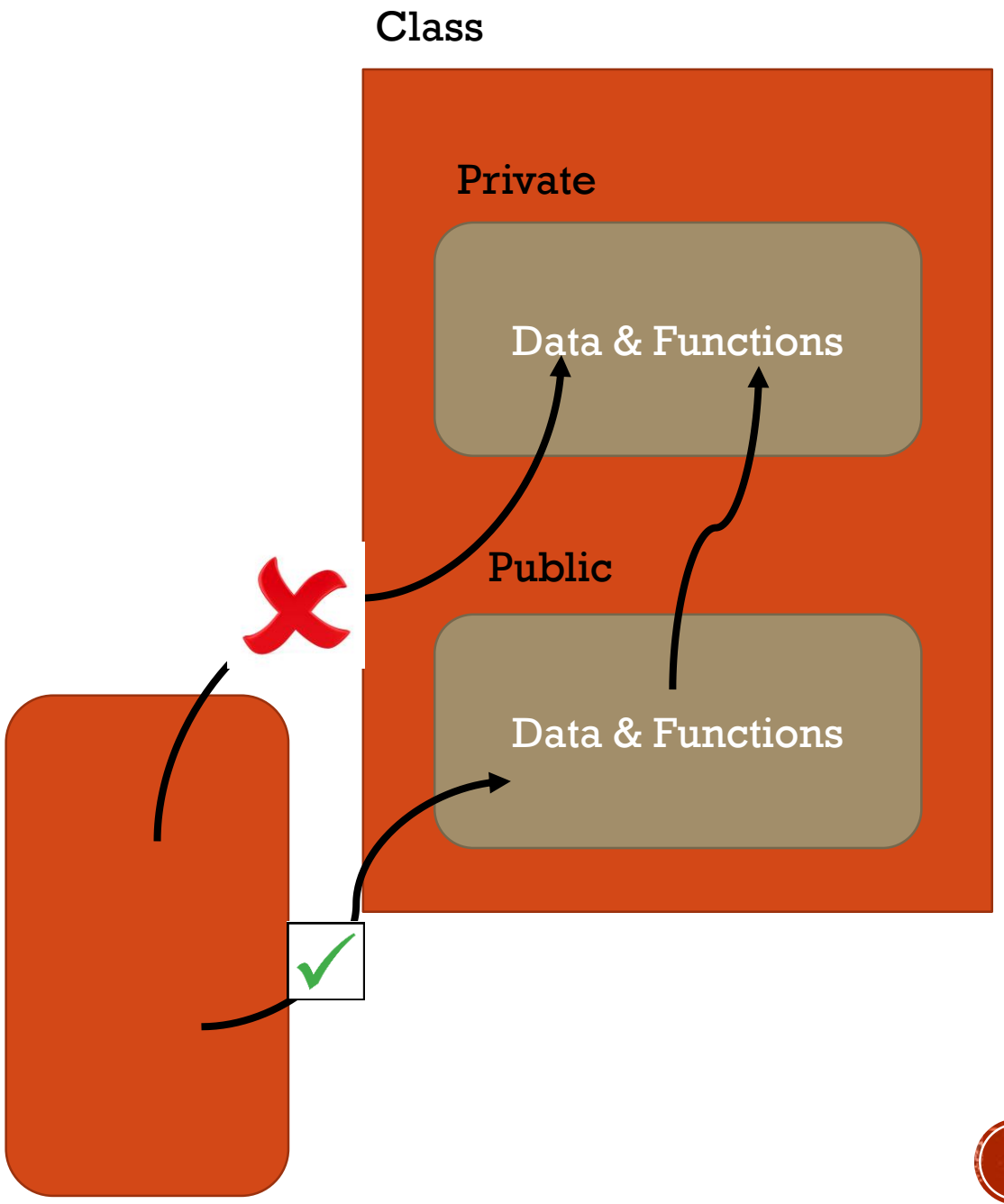


Compilation Error

The default access model for classes is private so all members after the class header and before the first member access specifier are private

ACCESS SPECIFIERS

- Define how the members (attributes and methods) of a class can be accessed.
- *Data hiding:* Key feature of object-oriented programming
  - The data is concealed/wrapped within a class so that it cannot be accessed mistakenly directly or by functions outside the class.
  - Private data or functions can only be accessed from within the class.
  - Public data or functions are accessible from outside the class.
- Hide data from parts of the program that don't need to access it.
- Data in one class is hidden from other classes.

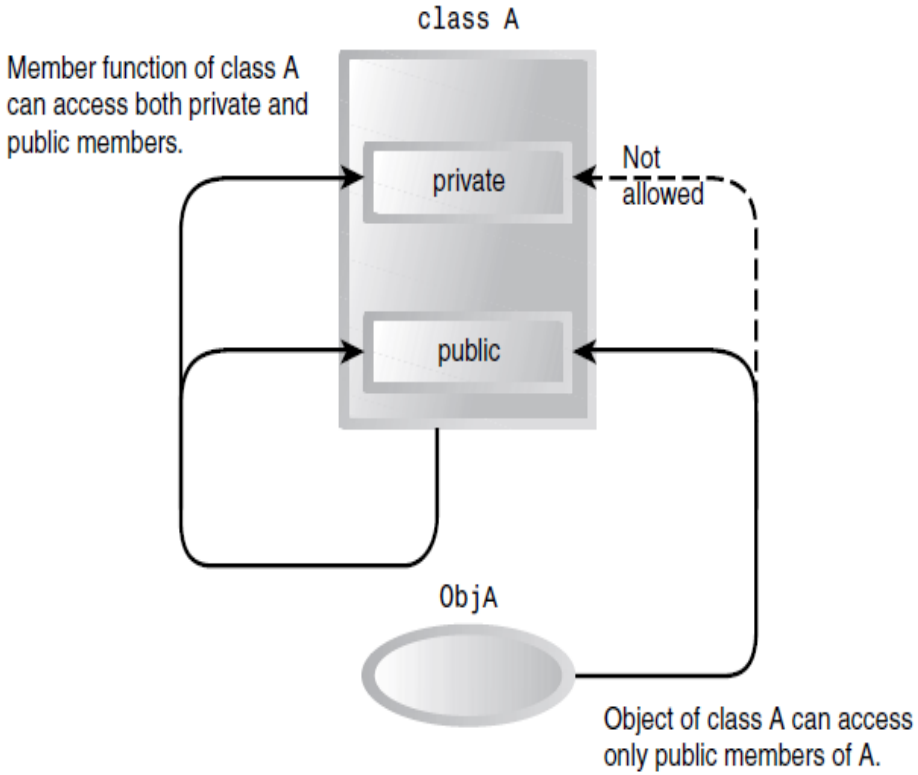


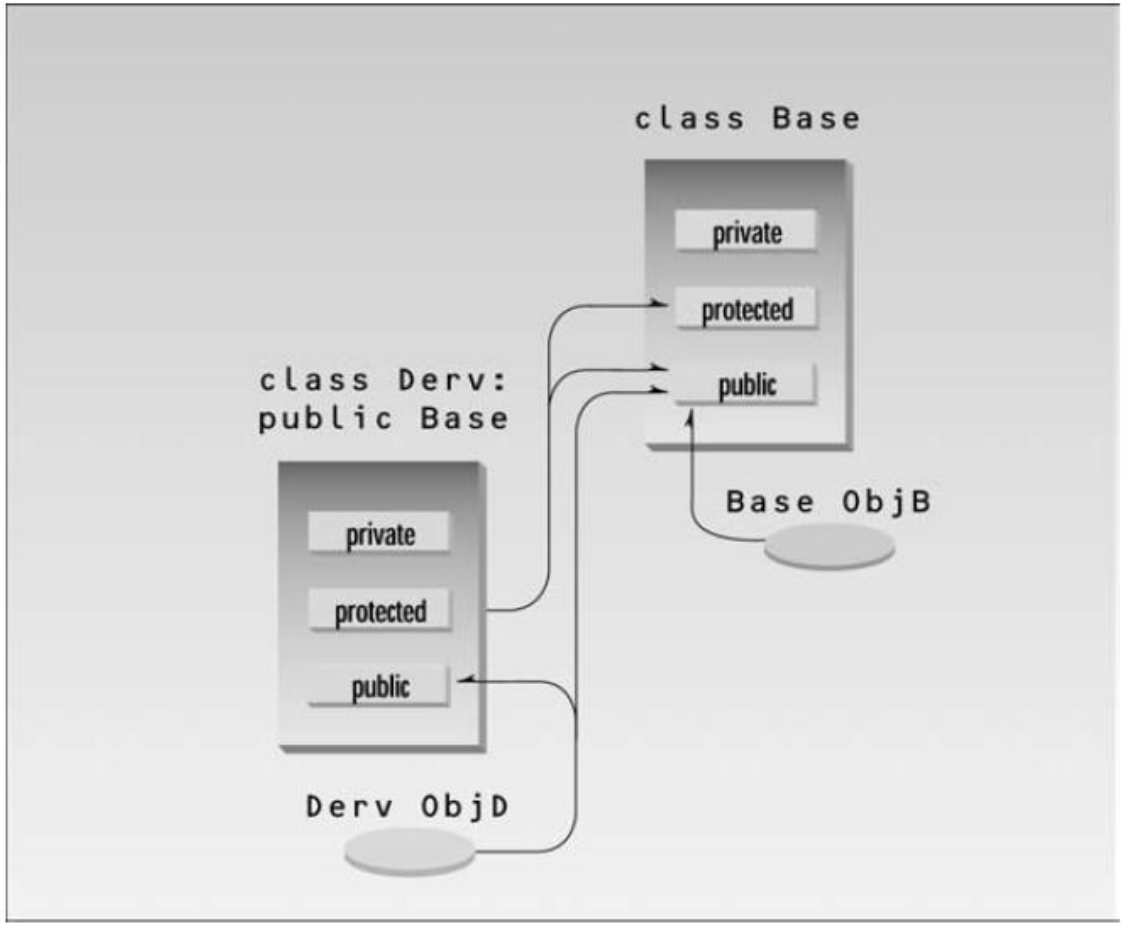
ACCESS SPECIFIERS

Keyword **public:** makes subsequent members public (can be accessed by non-member functions)

Keyword **private:** makes subsequent members private (cannot be accessed by non-member functions)

Keyword **protected:** Somewhat similar to the keyword private.


class A
Member function of class A can access both private and public members.
private
public
Not allowed
ObjA
Object of class A can access only public members of A.

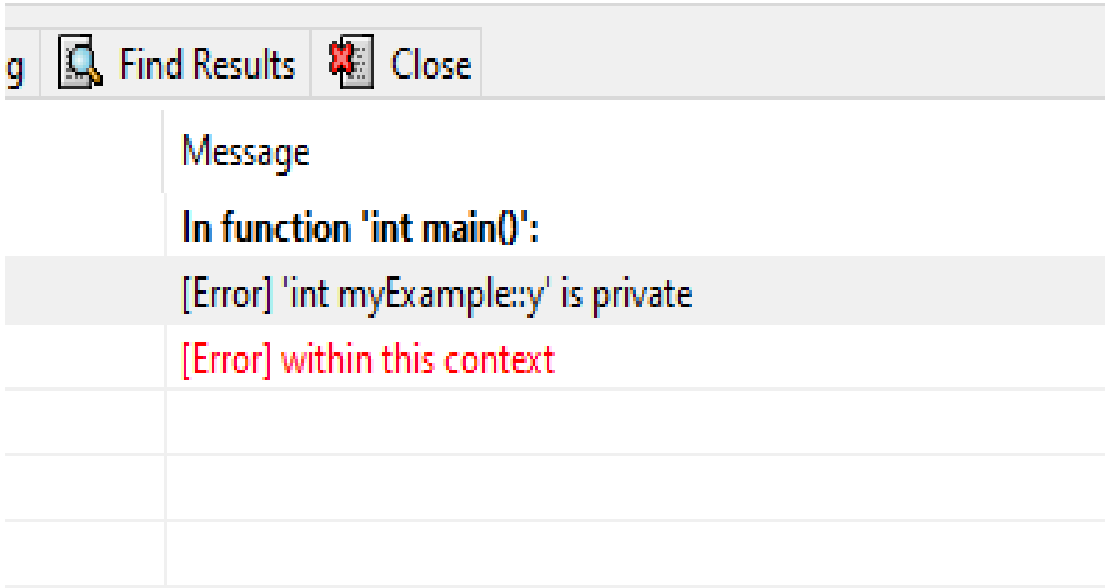

class Base
private
protected
public
Base ObjB
class Derv: public Base
private
protected
public
Derv ObjD

ACCESS SPECIFIERS-EXAMPLE

```
class myExample
{
  public:    // Public access specifier
    int x;   // Public attribute
  private:   // Private access specifier
    int y;   // Private attribute
};

int main() {
  myExample E1;
  E1.x = 3;
  E1.y = 4;
}
```

OUTPUT ?



DEFINING CLASSES

```
class ClassName{          //class Name
AccessSpecifier:          //public, private or protected
Datamembers;              //Variables/Attributes
MemberFunctions(){}       //Methods to access member functions/Behavior
//////////////////////////////////////////////////
AccessSpecifier:          //public, private or protected
Datamembers;              //Variables/Attributes
MemberFunctions(){}       //Methods to access member functions/Behavior
//////////////////////////////////////////////////
};                                  //end of the class Name
```

ACTIVITY

- Define a class to create a new datatype called Date (Should include day, month and year)
- Define two functions called
  - setDate to initialize values by passing parameters
  - getDate to print the value (Date) on the console
- Create a variable called "today" of data type Date
- Set the value to 5/June/2025
- Display the value on the console

CLASS: DATE

```
1   #include<iostream>
2   using namespace std;
3
4   class date
5   {
6       int day;
7       string month;
8       int year;
9
10      void setDate(int D,string M,int Y){
11          day = D;
12          month = M;
13          year = Y;
14      }
15
16      void getDate(){
17          cout<<"Day:"<<day<<" Month:"<<month<<" Year:"<<year<<endl;
18      }
19  }; // end of class definition
20
```

CLASS AS A DATA TYPE

- Once a class has been declared, it can be used as object/array declarations

Example:

```
Car carObj,                  // object of type Car
    Car carArray[10];        // array of Car objects
      Car *newCar;           // Object pointer
```

The class name becomes the new type specifier.

DATA MEMBERS AND MEMBER FUNCTIONS

Accessing Data members / Member functions

- DOT (.) Operator
  - Example: MyObject.variableX; MyObject.func();
- Refer to a member of a class via a pointer to an object, (->) Operator

A *pointer* variable, or simply a *pointer*, contains the location, or address in the memory, of a memory cell

ACTIVITY

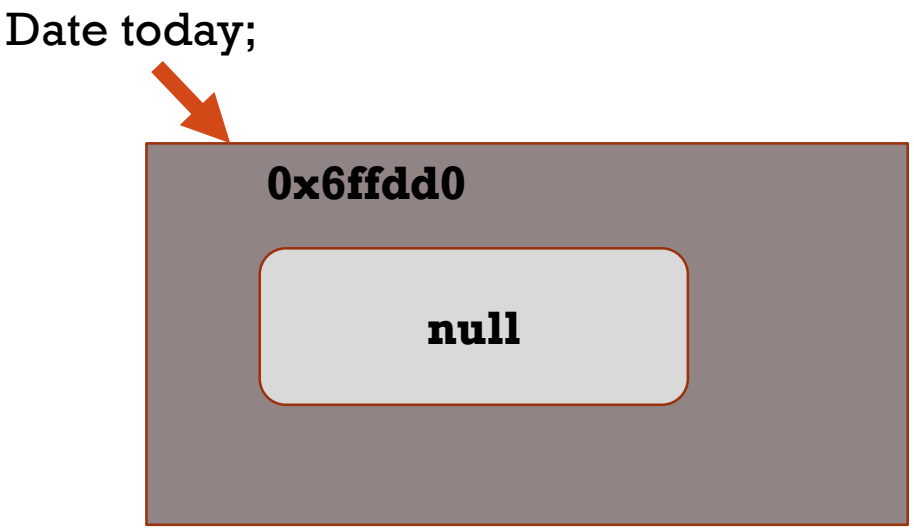
- Define a class to create a new datatype called Date (Should include day, month and year)
- Define two functions called
  - setDate to initialize values by passing parameters
  - getDate to print the value (Date) on the console
- Create a variable called "today" of data type Date
- Set the value to 28/November/2024
- Display the value on the console
- **Create a pointer variable called ptrDate to assign the value of "today"**
- **Display the value of ptrDate**

ACCESS OPERATOR (->): EXAMPLE

```
int main()
{
  date today;
  today.setDate(5,“June",2025);

            date *ptrDate;
            ptrDate=&today;

            cout<<"CHECK POINTER"<<endl;
            ptrDate->getDate();
}
```



DATA MEMBERS AND MEMBER FUNCTIONS

Accessing Data members / member functions

- DOT (.) Operator
  - Example: MyObject.variableX; MyObject.func();
- Refer to a member of a class via a pointer to an object, (->) Operator
- Accessing a data member depends on the access modifier of the data member.

Defining Member functions in classes

- Inside class definition
- Outside class definition
  - use the scope resolution :: operator along with class name and function name.

```
4  class Date{
5      int day;
6      string month;
7      int year;
8
9      public:
10     void setDate(int,string,int);
11     void getDate();
12 };
13
14 void Date::setDate(int d,string m,int y){
15     day=d;
16     month=m;
17     year=y;
18 }
19
20 void Date::getDate(){
21     cout<<"Day: "<<day<<" Month: "<<month<<" Year: "<<year<<endl;
22 }
```

DEFINING MEMBER FUNCTIONS

- The **scope resolution operator (::)** can be used to define member functions outside the class definition.
- Member functions defined inside a class definition are automatically treated as **inline functions**.
- Member functions that contain **more than one statement** should be defined outside their class definition.
- Members of a class have **class scope** – data members can be referred to directly inside the member functions from that class.

SCOPE RESOLUTION OPERATOR

::

Different Functions …

ACTIVITY

```
Value of the Variable a: 0
Calling from the Func Function:
Variable a is 100
Variable b is 67
```

```
1  #include<iostream>
2  using namespace std;
3
4   int a=3;
5
6  void func (int x) {
7       int b, a=100;
8           b=x;
9               cout<<"Calling from the Func Function: "<<endl;
10              cout<<"Variable a is "<<a<<endl;
11              cout<<"Variable b is "<<b<<endl;
12  }
13
14 int main()
15 {
16      int a=0;
17      cout<<"Value of the Variable a: " <<a<<endl;
18      func(67);
19 }
```

SCOPE OF VARIABLES

- The extent of the program code within which the variable can be accessed or declared or worked with.
- **Local Variables**
  - Variables defined within a function or block are local to those functions.
    - Declared inside a block.
  - Local variables **do not exist** outside the block
    - **Can not** be accessed or used outside that block.
- **Global Variables**
  - Can be accessed from any part of the program.
  - Available throughout the lifetime of a program.
  - Declared at the top of the program outside all of the functions or blocks.

SCOPE OF VARIABLES...

- Usually two variables with the same name are NOT allowed to be defined.
  - Within the SCOPE
- If the variables are defined in different scopes then the compiler allows it.
- If there is a local variable and a global variable with the same name
  - The **compiler gives precedence to the local variable**
- **How can you ACCESS the global variable???**
  - Use the **scope resolution operator**.
  - SCOPE RESOLUTION OPERATOR ■ ■ ■ ■

TO ACCESS THE GLOBAL VARIABLE

Scope Resolution Operator

```
1  #include<iostream>
2  using namespace std;
3
4   int a=3;
5
6   void func (int x)
7   {
8       int b, a=100;
9           b=x;
10              cout<<"Calling from the Func Function: "<<endl;
11              cout<<"Variable a is "<<a<<endl;
12              cout<<"Variable b is "<<b<<endl;
13  }
14
15 int main()
16 {
17      int a=0;
18
19      cout<<"Value of the Local Variable a in Main: " <<a<<endl;
20      cout<<"Value of the Global Variable a " <<::a<<endl;
21      func(67);
22 }
```

```
Value of the Local Variable a in Main: 0
Value of the Global Variable a 3
Calling from the Func Function:
Variable a is 100
Variable b is 67
```

ACTIVITY

ACTIVITY

```
40
3
X-a = 2

1
```

```cpp
#include<iostream>
using namespace std;

int a=3;

class X{
    int a;

    public:

    void setValue(int x){
        a=x;
    }

    int increment_a(){
        return (a++);
    }

    void printX(){
        cout<<"X-a = "<<a<<endl;}
}b;
```

```cpp
int main()
{
    int a=40;
    b.setValue(0);

    cout<<a<<endl;
    cout<<::a<<endl;

    a=b.increment_a();
    a=b.increment_a();
    b.printX();

    cout<<endl<<a<<endl;
}
```

SCOPE RESOLUTION OPERATOR

Used to

- **access** a **global variable** when there is a local variable with the same name,
- **define** a function **outside a class.**

Activity

- **access** elements having the same name inside **two namespaces**
  - Use the namespace name with the scope resolution operator to refer to that class

    using namespace std; & cout<< OR

    std::cout

TO ACCESS ELEMENTS HAVING THE SAME NAME INSIDE TWO NAMESPACES

Scope Resolution Operator

NAMESPACE

- A space where we can define or declare identifiers i.e. variables, methods, and classes.
- Can define Functions, classes, variables having the same name in different libraries.
- Syntax:

```
namespace  namespace_name
{
    // code declarations i.e. variable  (int a;)
    method (void add();)
    classes ( class student{};)
}
```

Calling the namespace

namespace_name: :code; // code could be variable , function or class.

ACTIVITY

```
Inside second_space
Inside first_space
```

```cpp
#include <iostream>
using namespace std;
// first name space
namespace first_space{
    void func(){
        cout << "Inside first_space" << endl;
    }
}

// second name space
namespace second_space{
    void func(){
        cout << "Inside second_space" << endl;
    }
}

using namespace second_space;
int main ()
{
    func();
    first_space::func();
}
```

SCOPE RESOLUTION OPERATOR

Used to

- **access** a **global variable** when there is a local variable with same name,
- **define** a function **outside a class**.
- **access** elements having the same name inside **two namespaces**
  - Use the namespace name with the scope resolution operator to refer to that class

    using namespace std; & cout<< OR

    std::cout
- **refer** to a **class inside** another **class**

ACTIVITY

```
1  // Use of scope resolution operator: class inside another class.
2  #include<iostream>
3  using namespace std;
4
5  class outside
6  {
7  public:
8      int x=0;
9
10     class inside
11     {
12     public:
13             int x = 1;
14             int y = 56;
15             int foo(){
16                 return x;
17             }
18     };
19 };
```

```
21 int main(){
22     outside A;
23     outside::inside B;
24     cout<<B.foo()<<endl;
25     cout<<B.y<<endl;
26 }
```

```
1
56
```

TO REFER TO A CLASS INSIDE ANOTHER CLASS

Scope Resolution Operator

ACTIVITY

```
#include<iostream>
using namespace std;

class outside
{
public:
    int x=0;

    class inside
    {
    public:
        int x = 1;
        int y = 56;
        int foo(){
            return x;
        }
    };
};

```

```
int main(){
    outside A;
    outside::inside B;
        cout<<B.foo()<<endl;
        cout<<B.y<<endl;
        cout<<A.y;
}

```

9 - ...	In function 'int main()':
Obj...	[Error] 'class outside' has no member named 'y'

ACTIVITY

```
 2  #include<iostream>
 3  using namespace std;
 4
 5  class outside
 6  {
 7  public:
 8      int x=0;
 9
10      class inside
11      {
12      public:
13              int x = 1;
14              static int y;
15              int foo(){
16                  return x;
17              }
18      };
19  };
```

```
22  outside C; //Global object
23
24  int outside::inside::y = 5;
25
26  int main(){
27
28      outside A;  //local object
29      outside::inside B;
30
31      cout<<B.foo()<<endl;
32      cout<<outside::inside::y<<endl;
33      cout<<B.y<<endl;
34      cout<<C.x<<endl;
35  }
```

```
1
5
5
0
```

SCOPE RESOLUTION OPERATOR

Used to

- **access** a **global variable** when there is a local variable with same name,
- **define** a function **outside a class**.
- **access** elements having the same name inside **two namespaces**
  - Use the namespace name with the scope resolution operator to refer to that class

    using namespace std; & cout<< OR

    std::cout
- **refer** to a **class inside** another **class**
- **access** a class's **static variables**.
- **access variables** with the same name in two ancestor classes in **multiple inheritance**.

ACTIVITY

```
1  #include<iostream>
2  using namespace std;
3
4  class Enclosing {
5  int x=0;
6  /* start of Nested class declaration */
7  class Nested {
8      int y=1;
9  }; // declaration Nested class ends here
10
11 void EnclosingFun(Nested *n) {
12         cout<<n->y;
13         cout<<x<<endl;
14         cout<<y<<endl;
15 }
16 }; // declaration Enclosing class ends here
17
18 int main()
19 {
20
21 }
```

DYNAMIC MEMORY ALLOCATION

DYNAMIC MEMORY ALLOCATION

- Static Memory allocation: Global and Local variables
  - The memory space is allocated by the compiler.
  - Done before the program executes.
- Can one explicitly allocate and deallocate space at *run-time?*
  - *YES / WHY?*
  - Advantages:
    - No need to know how much amount of memory is needed beforehand.
    - *Dynamically* allocate exactly the correct amount of space for the variables
    - Use the memory space more efficiently…etc.
  - Use a pre-defined keyword

    new to allocate storage and

    delete to deallocate the storage

DYNAMIC MEMORY ALLOCATION: *NEW/DELETE*

```
int main()
{
  // Allocated memory  dynamically.
  int *intPtr1 = new int;
  int *intArrayPtr2 = new int[10];

  // Dynamically allocated memory is deallocated
  delete intPtr1;
  delete [] intArrayPtr2;          //Don't forget the square brackets [] for an array
}
```

'new' allocates storage for the object and returns a pointer to the allocated storage

Good Coding Practise: Free dynamic memory when no longer required. Use the **delete** operator to release dynamic memory

DYNAMIC MEMORY ALLOCATION: *NEW/DELETE*

- Always check whether the memory has been successfully allocated

```
#include <cassert>
Time *timearrayPtr = new Time[10];
assert(timeArrayPtr != 0);
```

- **new** returns a null pointer if the allocation is unsuccessful